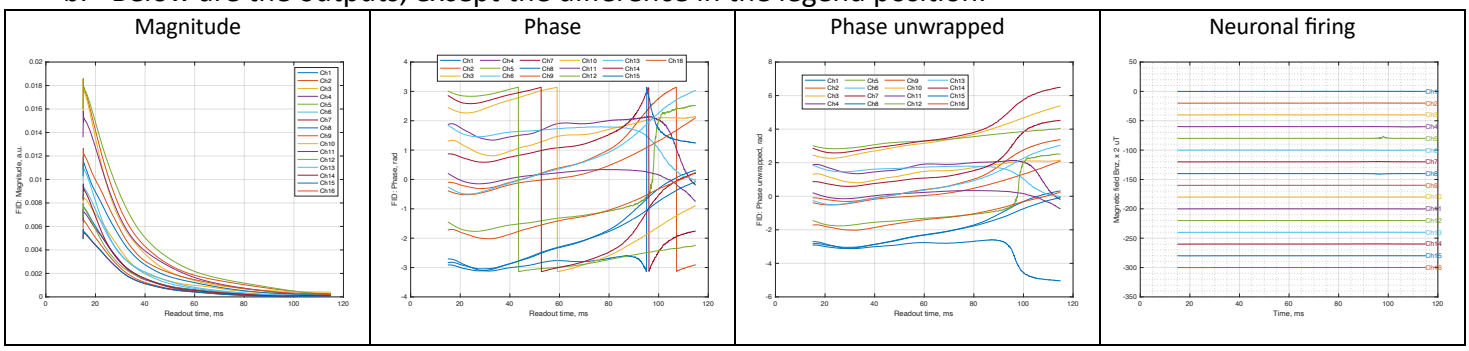
ReadMe (for qsMRI testing) *by Yongxian Qian, PhD, April 7, 2026.*

1. **Software required:** MATLAB version R2021a or later. You already have it on your computer.

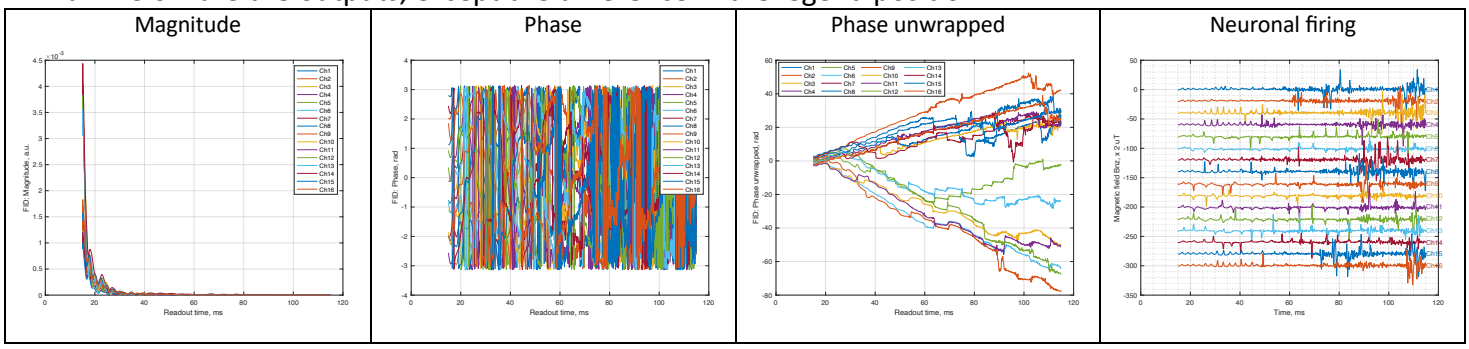
2. **Source code and data:** download from the sharing link provided by the authors, all the shared code and data into **one** folder on **your** computer for the testing.

3. **Simulation data (Extended Data Fig. 1):**
   a. To create the simulated FID data, run the code: *qsMRI_Simu_Data_Varify*. It produces *fidSimuData.mat*, which includes two data sets: *fidSimu* (no noise) and *fidSimuNoisy* (with noise at SNR=250).
   b. To calculate neuronal firing, run the code: *qsMRI_NeuroFiring_Prep_Simulation.*

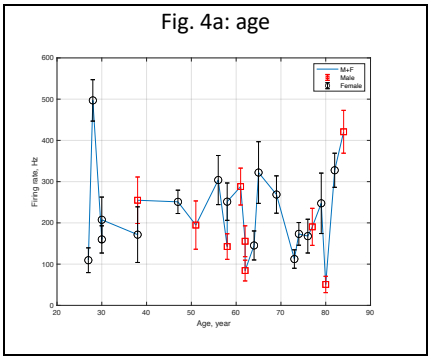
4. **Experimental data (Fig. 2, phantom data):**
   a. Run *qsMRI_NeuroFiring_Prep_Bottle*
   b. Below are the outputs, except the difference in the legend position.



5. **Experimental data (Fig. 3, 25-year-old male):**
   a. Run *qsMRI_NeuroFiring_Prep_H001*
   b. Below are the outputs, except the difference in the legend position



6. **Experimental data (Fig. 4a, firing rate with age):**
   a. Run *qsMRI_NeuroFiring_Prep_Age*
   b. Get the output the same as Fig. 4a. Other figures (4b-d) can be derived from the data for 4a.



7. **Experimental data (Fig. 5, firing rate under a task):**
   a. Run *qsMRI_NeuroFiring_Prep_AgeMotor*
   b. Get the outputs the same as Fig. 5c,e. Other figures (5d,f) can be derived from the data for 5c,e, respectively.

Fig. 5c: tasks

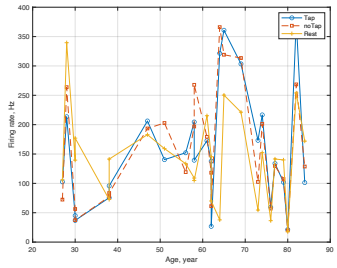
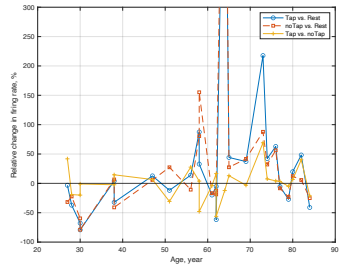


Fig. 5e: relative change



c. Run *qsMRI_NeuroFiring_Prep_H051v1*
d. Get the outputs the same as Fig. 5b.

Fig. 5b: Tap

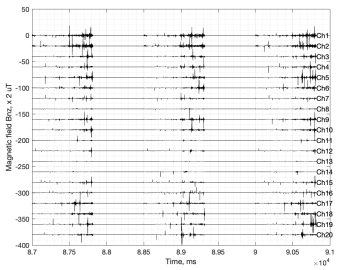


Fig. 5b: noTap

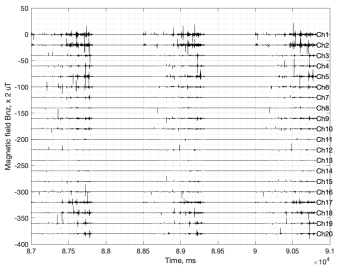
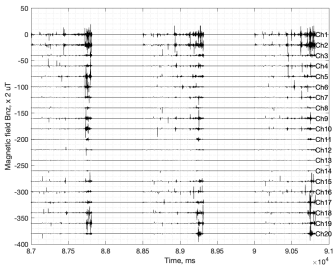


Fig. 5b: Rest



8. **Experimental data (Fig. 6, Epilepsy and HC):**
   a. Run *qsMRI_NeuroFiring_Prep_H015v1_EP*
   b. Get the outputs the same as Figs. 6a,d,e (Epilepsy-History), except vertical scales in d & e.

Fig. 6a: iFID #7

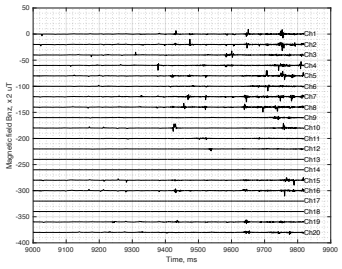


Fig. 6a: iFID #17

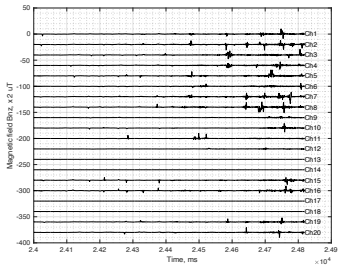


Fig. 6d: Firing rate

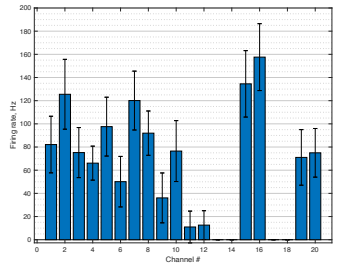
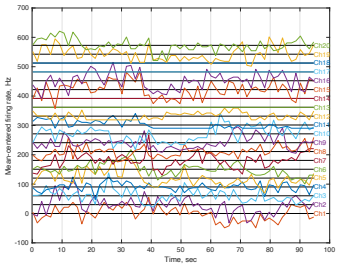


Fig. 6e: Firing dynamics



   c. Run *qsMRI_NeuroFiring_Prep_H014v1_HC*
   d. Get the outputs the same as Fig. 6b,f,g (matched Healthy Control), except vertical scales in f & g.

Fig. 6b: iFID #7

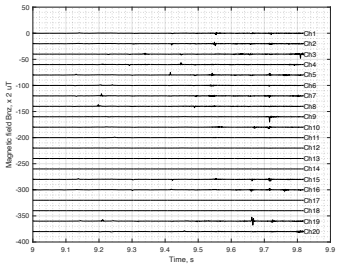


Fig. 6b: iFID #17

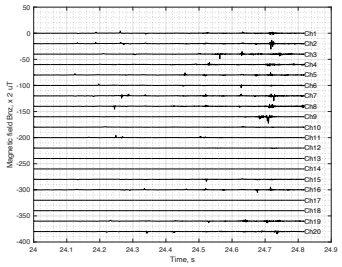


Fig. 6f: Firing rate

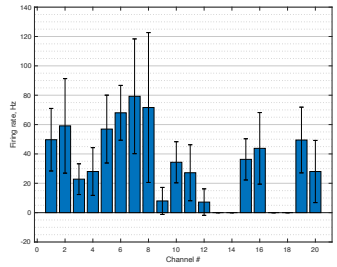
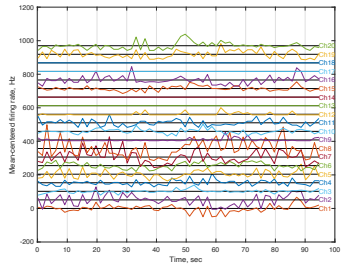


Fig. 6g: Firing dynamics



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